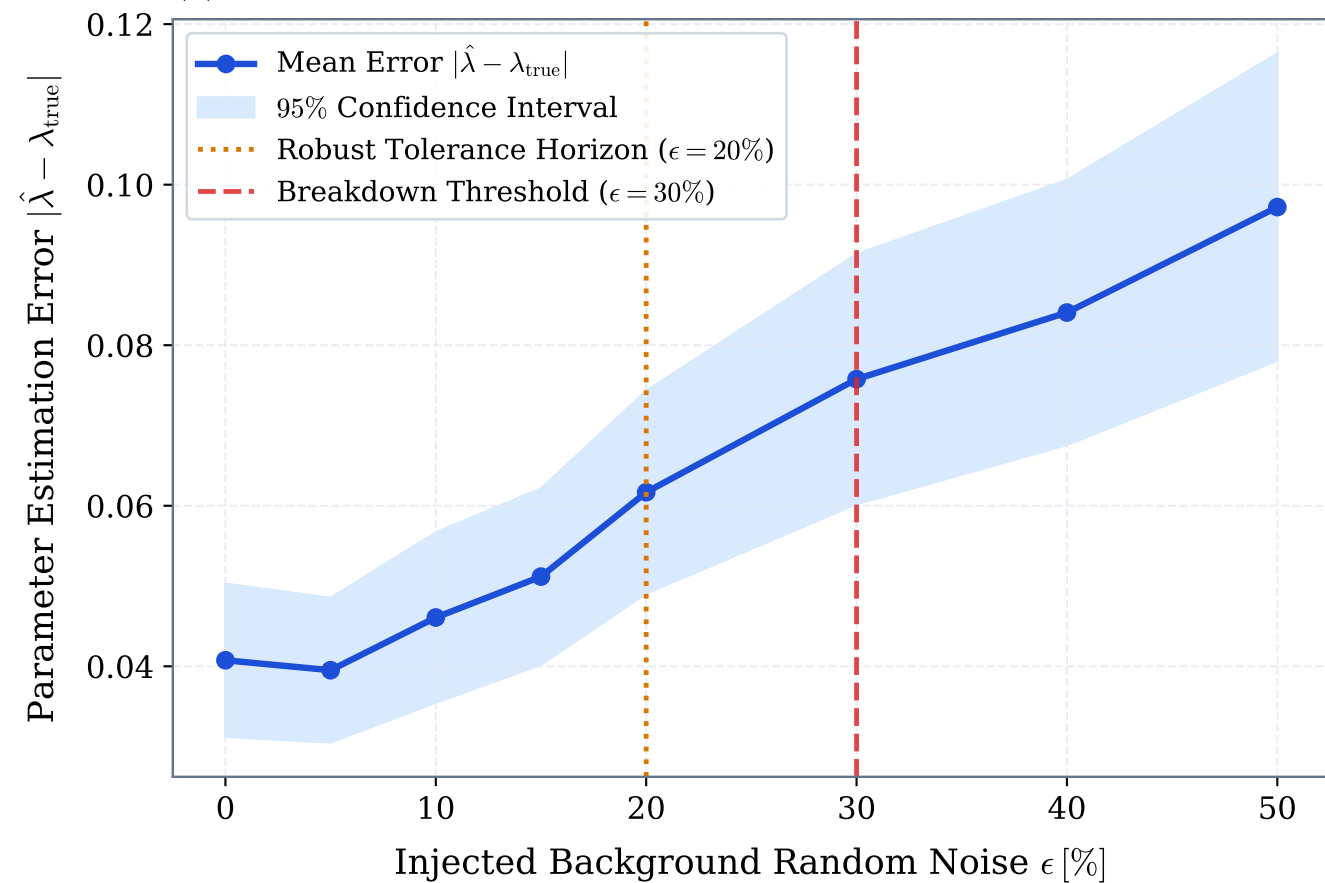
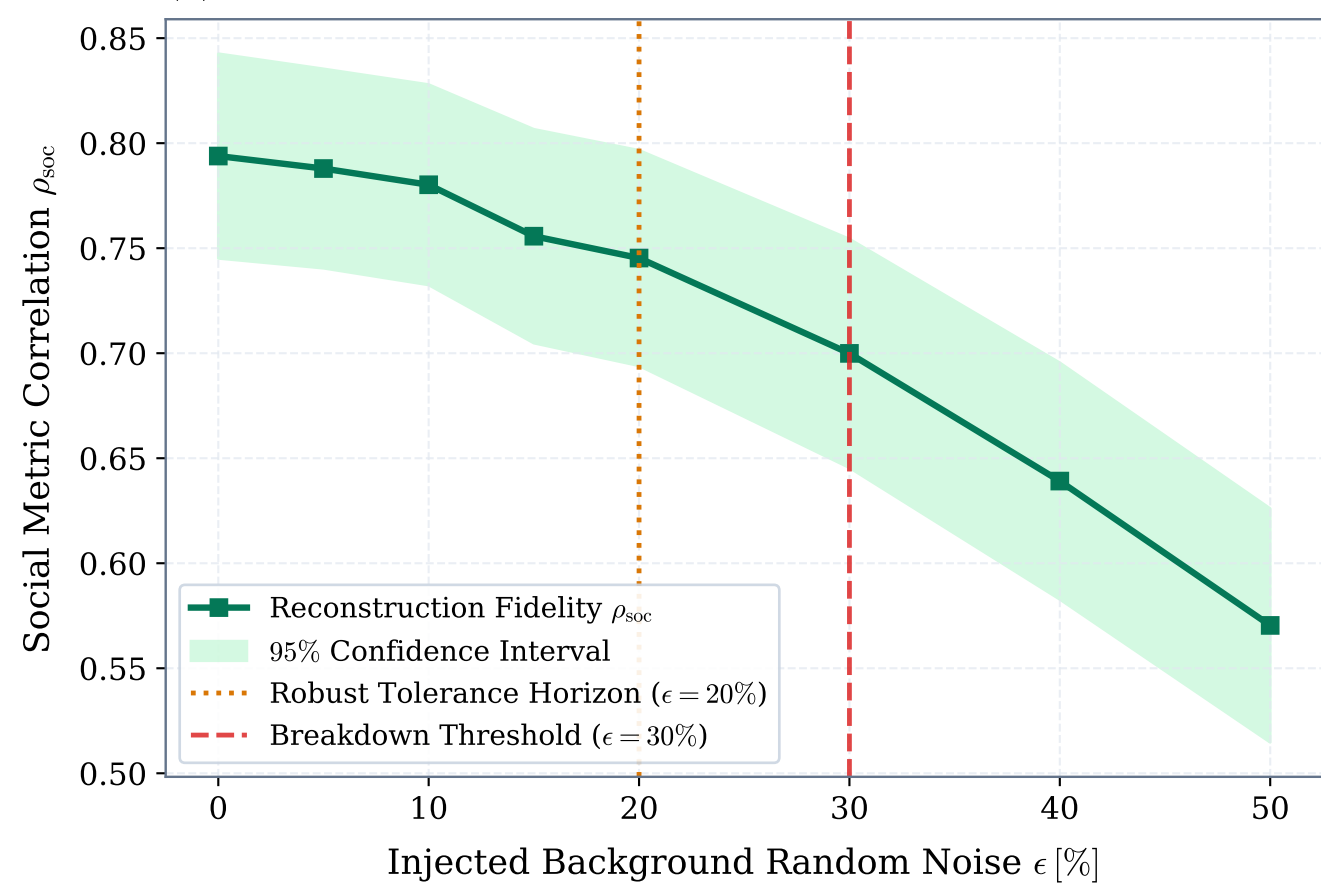


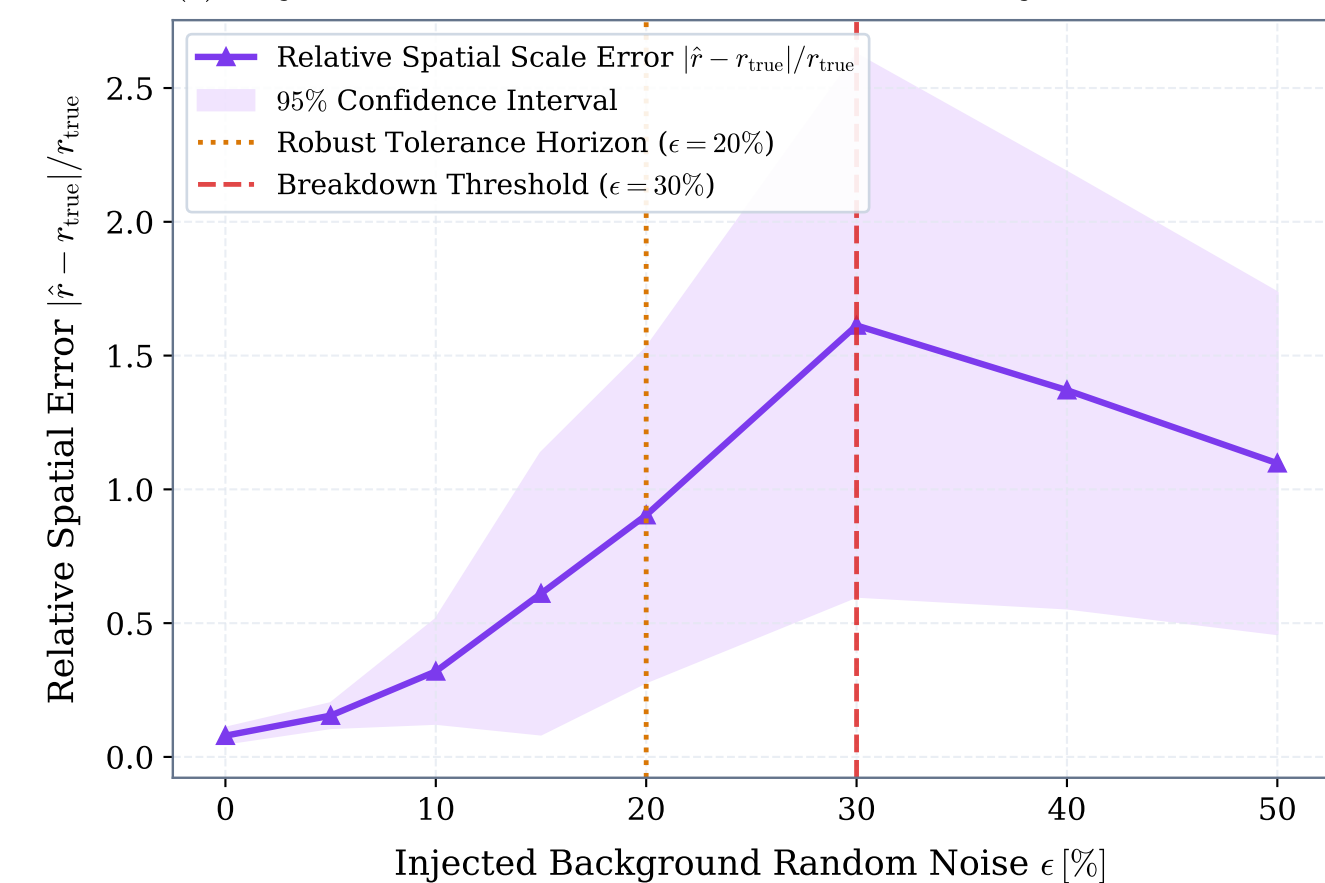
(a) Parameter Estimation Error vs. Noise Level



(b) Latent Social Space Recovery Fidelity



(c) Physical Interaction Reach Scale Recovery



(d) Normalized Pipeline Resilience under Noise

